COMP7507 Visualization and Visual Analytics

Project Proposal

Important Notes:

- The purpose of the proposal is to help you define the project scope and consolidate your ideas. Try your best to explore and investigate the feasibility of your ideas & plans when developing your proposal. You should better identify viable data sources so that it does not take you too much time on tedious data cleaning or processing.
- You need to write the proposal (i.e., fill in the text boxes) **using this template** and **submit one for each group**. Your inputs should be in font **Time New Roman 12pt black/auto.** Part B should contain no more than **5 pages**.
- You will receive feedback on your proposal from the teaching team within 1 week after submission, but you should continue to work on the project after proposal submission without idling.

Part A *(Basic Information)*

(i) Group Information

Group Number: 20

Group Members:

Name	*Student ID*
ZHOU YONGXI	3036790106
LU ZHOU	3036791370
ZHANG YUE	3036791148
WU JIELIN	3036793201
CHEN SHUFANG	3036808343

(ii) Project Title

Shanghai Resale Housing Price Patterns and Community Recommendation

Part B *(Proposed Method) - Maximum 5 pages*

(i) Project Objectives

This project will examine Shanghai resale asking-price patterns, compare district industrial context, test how metro access, nearby medical facilities and schools relate to community prices, and build an interactive Tableau tool that ranks communities within a selected district according to user-defined transport, medical and education preferences while displaying price and listing evidence.

(ii) Dataset Description

Original housing source. VNVILE's Shanghai Second-Hand House Dataset on Kaggle distributes property records collected from Fang.com as a Parquet file. The supplied extract contains 5,850 resale listings and 43 raw fields covering district, sub-district, community, asking price, estimated price, area, layout, building characteristics, listing date, coordinates and source URL. The raw file will be retained for lineage and reproducible checks: https://www.kaggle.com/datasets/vnvile/shanghai-second-hand-house-dataset.

Official district source. Shanghai Statistical Yearbook 2025 supplies 2024 observations for all 16 districts. The project will use district area, population and three economic measures: industrial gross output, construction gross output and operating revenue of service enterprises above designated size. Because these measures have different statistical scopes, they will be viewed separately and converted to intensity per square kilometre rather than added as industry shares: https://tjj.sh.gov.cn/tjnj/tjnj2025.htm.

Spatial and amenity source. AMap Web Service APIs provide community geocoding, the nearest metro station and walking route, and counts of hospitals and schools within 500 metres of each community centre. These fields measure proximity and facility counts, not medical quality, school quality or admission eligibility. AMap documentation: https://lbs.amap.com/api/.

Prepared Tableau inputs. Our group processed and transformed the source data to produce three Tableau-ready tables, which are documented in our GitHub repository. Version 2 contains Listing (5,850 rows), Community (2,242 rows, with complete metro, hospital and school fields), and District (16 rows). In Tableau, Listing will be related to Community using District and Community, while Community will be related to District using District: https://github.com/kkzyx/shanghai-house-visualization/tree/main/v2.

(iii) Methodology

(1) Tools/Approaches

Tableau will be used as the main tool for visualisation and interactive analysis, while Python will be used for data cleaning, organising geocoding results, calculating facility distances and generating the analysis tables. The final analysis will consist of three connected interactive dashboards.

Before the analysis, the data will be organised at the level required by each task:

- Listing data will be used to calculate the median housing price, number of listings and price distribution for each community.
- Metro data will be used to calculate the walking time from each community to the nearest metro station.
- Medical and education facility data will be used to calculate the number of hospitals and schools within a specified distance of each community.
- District-level industry data will be used to describe the industrial background of different areas, including indicators for industry, construction and services.
- The recommendation function will use the community as the unit of analysis and calculate transport, medical and education rankings within the district selected by the user.

The median housing price will be used as the main price measure because a small number of unusually expensive or inexpensive listings may substantially affect the mean, whereas the median better represents the typical asking price of a community. The number of listings will also be displayed to indicate when a community-level price is based on a relatively small sample.

(2) Justification

Dashboard 1: Geographic Location, Industrial Context and Housing Prices

This dashboard addresses the following question: Where are Shanghai's high- and low-price communities located, how do housing-price distributions differ among areas, and are these differences associated with the industrial background of each area?

Previous research on Shanghai has shown that housing prices exhibit clear spatial differences and are associated with accessibility and nearby amenities (Li et al., 2019).

1. Community housing-price map. Each mark represents a community. Its position is determined by the community's coordinates, colour represents median unit price, and mark size represents the number of listings. The map directly reveals spatial clustering, such as whether high-price communities are concentrated in central districts and whether lower-price communities are mainly located in peripheral areas.

2. District housing-price box plot. The box plot presents the median, interquartile range and outliers of community prices in each district. A comparison based only on district averages would hide within-district differences. The box plot therefore shows whether high- and low-price communities coexist within the same district and whether its price distribution is stable.

3. Scatter plot of housing prices and industry indicators. Each mark represents a district. The horizontal axis shows the industry indicator selected by the user, and the vertical axis shows the district's median community price. The available indicators are industrial output intensity, construction output intensity and service-sector operating-revenue intensity. A parameter allows the user to switch among the three indicators and observe how different industrial backgrounds are associated with district housing prices.

Industry indicators will be used only at district level as contextual information for explaining regional differences. The scatter plot is intended to identify patterns of association and will not treat an industry indicator as a direct cause of housing-price change.

The map is suitable for answering where prices are high or low; the box plot is suitable for showing how much prices vary within each district; and the scatter plot is suitable for examining whether district housing prices vary together with industrial context. The three views therefore address spatial distribution, price distribution and regional background rather than repeating the same information.

Dashboard 2: Transport, Medical Care and Education in Relation to Housing Prices

This dashboard addresses the following questions: How are transport, medical and education conditions around a community separately related to housing prices, and which relationship appears more evident?

Previous research has also found that proximity to metro stations and broader transit-network accessibility are associated with housing-price premiums in Shanghai (Liu et al., 2022).

The dashboard will contain three parallel modules with a consistent visual structure:

1. Transport and housing prices. The horizontal axis shows walking time from a community to the nearest metro station, the vertical axis shows the community's median unit price, and each mark represents one community. A trend line displays the overall relationship between walking time and price. Because a shorter walking time indicates better transport accessibility, this direction will be stated clearly in the title and tooltip.

2. Medical care and housing prices. The horizontal axis shows the number of hospitals within 500 metres of a community, the vertical axis shows the community's median unit price, and each mark represents one community. A trend line displays the overall relationship between the number of medical facilities and price. The result will not interpret hospital count as the quality of medical services.

3. Education and housing prices. The horizontal axis shows the number of schools within 500 metres of a community, the vertical axis shows the community's median unit price, and each mark represents one community. A trend line displays the overall relationship between the number of education facilities and price. School count represents proximity to facilities rather than school quality or school-district eligibility.

The three modules will use the same vertical scale, colour encoding and community filters so that users can compare transport, medical care and education directly. Mark size will represent the number of listings, and communities with small samples will use lower opacity so that a median based on only a few listings is not mistaken for a stable result.

When the user selects a district on the map or through a district filter, all three modules will update together. Selecting a community will highlight the same community across the other views, allowing the user to compare its performance in all three dimensions.

Three parallel scatter plots are suitable because the project must analyse three separate relationships between facility conditions and housing prices. Combining the indicators in one plot would mix different units, directions and meanings. The parallel design supports direct comparison in the final dashboard and also allows three team members to develop the transport, medical and education analyses separately under a common specification.

Dashboard 3: Community Recommendation Based on Personal Preferences

This dashboard converts the observations from the first two dashboards into an actionable decision process: after a user has selected a district and expressed different preferences for transport, medical care and education, which communities best match those preferences?

Community recommendation is more suitable than listing recommendation as the main output. Transport, medical and education indicators describe the surroundings shared by listings in the same community. Directly ranking listings could allow several listings from one community to occupy the top positions and make the results repetitive. The user will therefore identify a recommended community first and then inspect the available listings within it.

The user will complete the following process:

1. Select a district.
2. Set weights for transport, medical care and education.
3. Rank communities within the selected district.
4. Recalculate the composite score according to the selected weights.
5. Output the recommended communities.
6. Select a community and inspect its specific listings.

The three indicators will first be converted into scores with a consistent direction:

- A shorter metro walking time produces a higher transport score.
- A larger number of hospitals within 500 metres produces a higher medical score.
- A larger number of schools within 500 metres produces a higher education score.

To prevent the different units from affecting the calculation, the three indicators will each be converted to a score from 0 to 100 within the district selected by the user. The composite recommendation score will be calculated as: Composite Score = Transport Score × Transport Weight + Medical Score × Medical Weight + Education Score × Education Weight. The three weights will first be divided by their total, so a comparable score can still be calculated when the entered weights do not sum to 100. If an indicator has no variation within the selected district, every community will receive the same middle score for that indicator to avoid division by zero.

The recommendation dashboard will contain:

1. Community composite-score map. Map colour represents the composite score so that the user can see the locations of high-scoring communities within the selected district.

2. Community recommendation ranking table. The table presents community name, composite score, transport score, medical score, education score, median unit price and number of listings. It allows the user to see both the ranking and the reasons behind it.

3. Community indicator comparison view. After the user selects a community, the view shows its three component scores and compares them with the median level of the selected district. This explains the recommendation instead of presenting an unexplained total score.

4. Listing-detail table. After a community has been selected, the table lists its available properties, including total asking price, unit price, area, layout and listing link. Listing details are a subsequent query result and do not participate in the community composite-score calculation.

Weight parameters fit this objective because different users assign different importance to transport, medical care and education. Dynamic ranking reveals how a change in preferences changes the recommendation and turns the dashboard from a static description into a personal decision tool.

(3) Step-by-step Approach

The project will follow the sequence of identifying differences, examining relationships and supporting a choice.

Step 1. Define the unit of analysis and prepare the data. Duplicate records, missing prices, invalid areas and abnormal coordinates will be cleaned, and district and community names will be standardised. The median unit price, listing count, coordinates, nearest-metro walking time, hospital count within 500 metres and school count within 500 metres will then be calculated for each community. District industry data will be linked by district and used only for the regional-context analysis in Dashboard 1. Listing-level records will be retained for the final drill-down in Dashboard 3.

Step 2. Show the spatial differences in Shanghai housing prices. The user will first observe the spatial distribution of high- and low-price communities on the community price map and then use the district box plot to examine price variation within each district. After selecting a district, the map and box plot will update together. The user can then switch among industry, construction and service indicators and observe how district industrial background is associated with median housing price. This stage establishes the spatial context: housing prices are not evenly distributed across Shanghai, and substantial differences may exist both among districts and within the same district.

Step 3. Analyse the relationships between living facilities and housing prices. After understanding the price distribution, the user will enter Dashboard 2 and separately examine the relationships between housing prices and transport, medical care and education. The three views use a consistent visual structure to compare whether prices tend to change as metro walking time increases, as hospital count increases or as school count increases; whether these relationships are consistent across districts; and whether the observations are affected by small community samples or outliers. This stage identifies observable associations and will not claim that metro stations, hospitals or schools directly cause changes in housing prices.

Step 4. Convert the analysis into community recommendations. In Dashboard 3, the user will select a district and set three preference weights. The system will compare only communities within that district, avoiding direct competition between central and peripheral districts whose overall prices and resources differ. It will calculate the three standardised scores and then the composite score and within-district ranking. The map will show the locations of high-scoring communities, the ranking table will present the order, and the comparison view will explain each community's strengths and weaknesses. If the user increases the transport weight, communities closer to metro stations should rise in rank; if the education weight increases, communities with more nearby schools should rise in rank. These changes will be used to verify that the recommendation responds correctly to the user's preferences.

Step 5. Drill down from community recommendation to listing search. After selecting a recommended community, the user will see the listings available in that community, including price, area and layout, and can determine whether it contains options that also meet the user's budget and housing requirements.

The complete story is therefore: Where are Shanghai's housing prices high or low? → How large are the differences within each district? → How is industrial background associated with district housing prices? → How are transport, medical care and education separately related to housing prices? → Under a selected district and personal preferences, which communities are more suitable? → Which specific listings are available within a recommended community?

The three dashboards therefore perform the respective roles of spatial discovery, relationship analysis and decision support. They share district, community, price and facility measures, creating a continuous and complete analytical story.

Part C *(List of Tasks)*

(iv) Task List with Timeline and Division of Labour

The five members have separate primary ownership so that each person is responsible for a complete and identifiable output. Members 1-3 develop the three parallel modules in Dashboard 2 under the same visual specification; Member 4 develops Dashboard 1; and Member 5 develops Dashboard 3 and integrates the workbook. Member labels will be replaced by names before submission.

Timeline and shared deliverables

Period	Task and concrete deliverable	Lead	Support / review
4-9 Oct	Freeze the three-dashboard story, field definitions, visual specification, worksheet inventory and wireframes. Record the data grain and interpretation limit for every measure.	All members	Member 5 coordinates
10-18 Oct	Prepare and validate workstream data: metro and walking time (M1), hospital POIs (M2), school POIs (M3), housing and district industry data (M4), and scoring specification plus Tableau relationship audit (M5). Deliver one checked input table and data note per workstream.	Members 1-5	Paired cross-checks
19-30 Oct	Build the first working versions in parallel: transport, medical and education worksheets (M1-M3), Dashboard 1 (M4), and the Dashboard 3 scoring and ranking prototype (M5).	Individual owners	Assigned reviewers
31 Oct-9 Nov	Combine the three Dashboard 2 modules with the same price scale, district filter, listing-count treatment, colours, tooltips and linked highlighting. Complete Dashboard 3's map, ranking table, profile and listing drill-down.	M1-M3; M5	M4 checks visual consistency
10-17 Nov	Connect the three dashboards through navigation, filter actions and reset controls. Verify that district filtering occurs before ranking and that changing each weight	Member 5	All members test their workstreams

	changes the expected communities' positions.		
18-23 Nov	Run data, calculation and usability QA: row-count reconciliation, missing values, small-sample modes, outliers, ties, zero weights, no-variation cases and Jinshan exceptions. Prepare and rehearse the live walkthrough.	All members	M4 visual QA; M5 calculation QA
21 and/or 28 Nov	Deliver the live Tableau walkthrough. Each member presents the workstream they own and records feedback and required corrections.	All members	Member 1 records issues
24-29 Nov	Complete the Tableau package and group report. M1-M3 write their facility-price analyses, M4 writes the dataset and spatial-industry analysis, and M5 writes the recommendation method and integration. Exchange sections for final editing and reference checks.	Individual owners	All members cross-review
30 Nov	Run the submission checklist and submit the packaged workbook and the single combined PDF report.	Member 1	All members approve final files

Part D *(References)*

AutoNavi Software Co., Ltd. (n.d.). *AMap Open Platform Web Service API* [Computer software]. https://lbs.amap.com/api/webservice/

kkzyx. (2026). *Shanghai house visualization* (Version 2) [Data set]. GitHub. https://github.com/kkzyx/shanghai-house-visualization/tree/main/v2

Li, H., Wei, Y. D., Wu, Y., & Tian, G. (2019). Analyzing housing prices in Shanghai with open data: Amenity, accessibility and urban structure. *Cities, 91*, 165-179. https://doi.org/10.1016/j.cities.2018.11.016

Liu, Z., Li, Y., & Zhang, M. (2022). Transit network effects and multilevel access premiums: Evidence from the housing market of Shanghai, China. *Cities, 129*, Article 103841. https://doi.org/10.1016/j.cities.2022.103841

Shanghai Municipal Statistics Bureau. (2025). *Shanghai statistical yearbook 2025*. https://tjj.sh.gov.cn/tjnj/tjnj2025.htm

VNVILE. (2025). *Shanghai second-hand house dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/vnvile/shanghai-second-hand-house-dataset